in low light conditions.

The newer lidar sensors are now capable of detecting and measuring light levels as low as a few photons, which enables them to operate in near-total darkness. Another factor is the development of systems that use multiple sensors and data sources to create a more comprehensive view of the environment.

Some lidar manufacturers are developing algorithms and software that can compensate for low light conditions and improve the accuracy of measurements. While, overall, lidar systems still rely on some level of ambient light to operate, newer systems are becoming more capable of working in dark environments, which makes them more suitable for applications such as autonomous vehicles, surveillance, and search and rescue operations.

DSP based lidar

A lot of new lidar systems employ a digital dignal processor (DSP) to process the signals received by the sensor. ConnX DSP and Cepton Vista-X120 are examples of DSP based lidars.

In a DSP based lidar system, the DSP is used to perform real-time signal processing on the data received by the lidar sensor. The use of a DSP in a lidar system offers several benefits, such as enabling real-time processing of lidar data, which is essential for applications like autonomous vehicles and robotics. The DSP can quickly and efficiently analyse the data received by the lidar sensor and make decisions based on that data, without the need for human intervention.

DSP based lidar systems can be more accurate and reliable than other types of lidar systems. The DSP can perform complex calculations and signal processing tasks that are difficult or impossible to achieve with traditional analogue or digital signal processing techniques. This can lead to higher-quality lidar data and more accurate measurements of distance and other environmental factors.

Digital beam steering

Digital beam steering is a technique used in lidar systems to control the direction of the laser beam emitted by the sensor. Newly launched lidars, such as LeddarSteer and Tensilica ConnX DSPs, employ digital beam steering.

In this technique a laser beam is directed towards a target, and the reflected light is detected by a receiver to determine the distance and other characteristics of the target. Digital beam steering allows the sensor to control the direction of the laser beam, which can be used to scan the environment and build a 3D map of the surrounding area.

Digital beam steering enables the sensor to scan the environment very quickly and efficiently. By rapidly scanning the laser beam in different directions, the lidar system can capture a large amount of data in a very short amount of time, which is essential for applications like autonomous vehicles and robotics.

Software definable lidar

Lidars are used in a wide range of applications, each with different requirements and operating conditions. For example, a lidar system used for autonomous vehicles may need to be optimised for detecting small objects at long distances, while a lidar system used for industrial robotics may need to be optimised for detecting reflective surfaces in a cluttered environment.

So, unlike the traditional lidar systems, which had fixed hardware components that could not be easily modified or updated, the companies are now making software definable lidars. These lidar systems can be easily reconfigured and optimised using the software. By adjusting parameters such as laser power, scanning pattern, and detection sensitivity, a lidar system can be optimised for specific applications, enabling it to achieve higher performance and accuracy.

A great advantage of the software-defined lidars is that they can be easily updated and improved over time. As new software algorithms are developed, or new hardware components become available, they can remain up-to-date and competitive in a rapidly changing market. Some software-defined lidars that were launched recently are Leddarsteer, Vista-X120, and PreAct T30P.

The future of lidars

Lidar technology is used in a wide variety of applications and their potential use cases are constantly expanding. Many experts believe that lidars will become faster, higher powered with better range, precise, robust, and more economical.

The advancement in lidar technology, which includes improvement in hardware, integration of other sensors, and use of machine learning and artificial intelligence in the processing of data, will not only enhance the quality of current applications but will open doors for a lot of newer applications.

Lidars can become mainstream in healthcare to create 3D models of patient anatomy for surgical planning, or to monitor patient movement and mobility. They can also be used extensively in search and rescue operations, or be employed in future cities to monitor traffic patterns and optimise traffic flow, or to detect and monitor pollution levels in the air and water.

So, we have to wait and watch to know what the future really holds for lidars. EFY

The author, Sharad Bhowmick, works as a Technology Journalist at EFY. He is passionate about power electronics and energy storage technologies. He wants to help achieve the goal of a carbon neutral world